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Abstract

Mobile phones have evolved to be smart computers (smartphones) supporting a wide range of information services that can be accessed anytime and from (almost) anywhere. With the increasing number of users and greater incursion into people's life, smartphones have the potential to significantly influence the touristic experience. This study explores the mediation mechanisms of smartphones by examining stories provided by travelers related to their use of smartphones (and associated applications) for traveling purposes. The results reveal that smartphones can change tourists' behavior and emotional states by addressing a wide variety of information needs; in particular, the instant information support of smartphones enables tourists to more effectively solve problems, share experiences, and "store" memories. The implications of these findings are important in that they suggest a huge potential for smartphones in changing many aspects of the tourism business.

Keywords

smartphone, mobile communication, touristic experience, Internet, mediation

Introduction

Mobile phones have evolved in the past few years such that they have a variety of input capabilities and large screens, they provide reliable and unlimited Internet access, and they incorporate powerful location awareness functions (Want 2009). Importantly, the increased capabilities of mobile phones (i.e., smartphones) support thousands of mobile applications (apps), which extend the functionality of mobile phones to a wide range of information services such as specialized information search, social networking, navigation, etc. Indeed, recent studies indicate that smartphones and their apps have the potential to assist travelers by providing easy access to information anytime and (almost) anywhere (Brown and Chalmers 2003; O'Brien and Burmeister 2003; Rasinger, Fuchs, and Hopken 2007). Further, Kramer et al. (2007) found that travelers' activities can be easily changed by the use of smartphones. Saari, Yoo, and Tussyadiah (2008), Tussyadiah and Fesenmaier (2009), and Wang, Park, and Fesenmaier (2010) found that smartphones can mediate both the behavioral and psychological dimensions of the touristic experience by facilitating information search, information processing, and information sharing, by enabling a traveler to learn about new travel opportunities and to get to know better a destination, and by sharing photos and other "social" activities at any time during the trip.

To further understand the role of smartphones in mediating the touristic experience, this study examines tourists' stories about the use of smartphones in assisting their travel. The rest of this article is organized into four sections. The

next section provides a brief overview of the literature describing the conceptual foundation of the touristic experience and discusses the relationships among smartphones, tourist information needs, and touristic experience. The sampling and data analysis methods are described in the following section. In the results section, the mediation mechanisms are identified and illustrated in detail using ten interpretive maps of tourists' stories describing their use of travel-related smartphone apps. Last, the implications of the results are discussed within the context of future research and the marketing of tourism destinations.

Literature Review

Travel is a process in which tourists leave the place where they live and travel to different places, interact with the objects and people in those places, and document travel memories in the formats of photos and videos (Clawson 1963; Cohen 1979; Tussyadiah and Fesenmaier 2008). Some scholars suggest that travel is a "linear" process and have defined the touristic experience from a temporal perspective (Graburn 1989; Craig-Smith and French 1994; Jennings

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1997, 2006) which involves three phases: (1) the anticipatory phase; (2) the experiential phase; and (3) the reflective phase (Clawson 1963; Craig-Smith and French 1994). Within these phases, tourists complete a series of activities including information search, planning, making reservations, visiting, shopping, dining, and reflection. Besides the core activities of tourism, social and environmental activities have been recognized to be important components of the overall touristic experience (Andereck et al. 2006), and therefore from the temporal perspective, the touristic experience is best described as an “activity-based” process.

Considering that experiences are reflective and inherently personal (Holbrook and Hirschman 1982; Pine and Gilmore 1999), some scholars argue that the touristic experience lies “beyond” the temporal dimension (Jennings et al. 2009). For example, Urry (1990) introduced the concept of tourist gaze in arguing that the tourist subjectively objectifies and interprets the places visited. Uriely (2005) echoed this viewpoint by suggesting further that the diverse characteristics of the touristic experience can be explained by the diversity of tourist motivations. Botterill and Crompton (1996) also explored the touristic experience based on personal construct theory (PCT) and concluded that tourists have “idiosyncratic ways of seeing” (p. 77) in that the touristic experience cannot be separated from travelers’ psychological processes and emotional states.

The literature also indicates that travel is a sense-making process whereby travelers construct the touristic experience by learning, understanding, and feeling the places visited and the culture embedded in these places (Jennings and Weiler 2006). The places visited and cultures experienced are, in turn, connected to tourists by stakeholders including tourists, tourist providers, governments, communities and indigenous groups. As such, the stakeholders mediate the touristic experience by (mis)representing and/or being part of the tourism context. One of the well-known examples of mediators in the tourism setting is the tour guide; indeed, Cohen (1985) argues that contemporary tour guides serve an important mediatory role in that they often link tourists to locals (social mediation) and translate “the strangeness of a foreign culture into a cultural idiom familiar to the visitors” (p. 15) (cultural brokerage).

The development of mass media and technology enriches the meaning of mediation in tourism context (Gretzel et al. 2011). Following Leiper (1990), tourists first get connected with destinations/ attractions through messages circulated by the media such as film, TV, literature, magazines, and videos, which direct “tourists gaze” by constructing and reinforcing tourists’ anticipation to visit places (Urry 1990). Based on the concept of spatial appropriation, which is related to people’s mobility in different spaces, Jansson (2002, p. 435) organized media into a three-dimension mediascape including symbiotic mode (i.e. documentaries, photography), antagonistic mode (i.e. tourism brochures, souvenirs), and contextual mode (i.e. sport programs, movies). He argued

that different modes of mediascape induce a diversity of travel modes in the social and physical landscapes. Thus, travel becomes “a hermeneutic circle” (Lagerkvist 2008, p. 349) in which tourists make plans according to messages communicated by the media.

Recent studies have begun to examine the extent to which Internet-based systems mediate the touristic experience (Cheverst et al. 2000; Wang, Yu, and Fesenmaier 2002; Gretzel, Fesenmaier, and O’Leary 2006; Tussyadiah and Fesenmaier 2008, 2009). For example, Tussyadiah and Fesenmaier (2009) studied the role of online shared videos in mediating the touristic experience by providing mental pleasure through stimulating fantasies and daydreams and by providing access to foreign landscapes and socioscapes with narrative transportation. They describe the mediation mechanism from the relationship between imaginative hedonism and optimal experience where they argued that “the new media as a form of cultural artifact and the potential actions inherent in it can provide a certain type of pleasurable feedback to the investment of attention, cause a certain degree of involvement and generate an enjoyable experience” (p. 27). Additionally, mobile tour guides such as CRUMPET (Poslad et al. 2001), INTRIGUE (Ardissone et al. 2003), and P-Tour (Maruyama et al. 2004) have been designed to direct tourists by providing recommendations and visualizing the surroundings of tourists. Thus, mobile tour guides enhance the “mediated gaze” (Lagerkvist 2008, p. 351) by their highly interactive and personalization capabilities. As such, these studies suggest we need to better understand the relationship between information needs, information tools (i.e. the Internet, Smartphones, and their applications) and the touristic experience.

Smartphones as one kind of new media now can provide a wide range of information services to support not only basic travel activities such as planning, reservation, and navigation, but many “micro-moments” within the travel process such as finding gas stations, estimating waiting time of rides, and “seeing” places as they once were (Wang, Park, and Fesenmaier 2010). Thus, travelers’ information needs can be addressed within any stage of the travel process including anticipatory phase, experiential phase, and reflective phase (Gretzel, Fesenmaier, and O’Leary 2006). For example within the experiential phase (i.e., during the trip), smartphones provide access to location-based services (i.e., “destination guide” apps that can recognize the current location of tourists) and provide relevant suggestions based on tourists’ inquiries including restaurants, souvenir shops, gas stations and even a restroom. Thus, tourists may divert from their current route and initiate unplanned activities (Hwang 2011). Also, some “entertainment” apps enable tourists to plot photos onto a map immediately after photo-taking in order to share these photos with social networking websites. In this way, travelers can keep their friends up-to-date and enjoy feedback about their on-going travel experience. These capabilities suggest that smartphones have the potential to

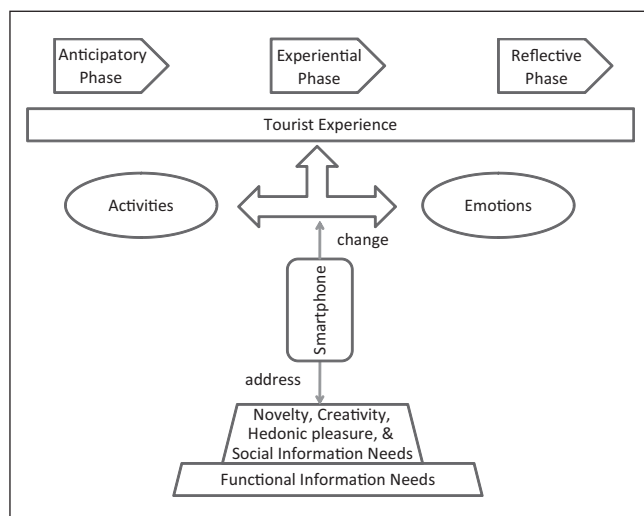


Figure 1. Mediation mechanisms of smartphones in the touristic experience

construct a very strong “mediated gaze” (Lagerkvist 2008, p. 351) in that they can shape traveler’s experiences by sharing stories and other forms of shared experiences (Gretzel 2010) (see Figure 1).

Method

Smartphones providers such as Apple, Blackberry, Verizon, AT&T, etc. offer access to a hundreds of thousands of smartphone apps through a variety of online systems including “iTunes” for iPhones and “Market” for Android-system supported phones. In customer evaluations/reviews of the smartphone apps, many users describe how the apps they purchased (or downloaded for free) affect their travel experience. This study uses these customer reviews to examine the nature of how smartphones mediate the touristic experience. Specifically, the customer reviews used in this study are associated with the 100 most popular smartphone travel-related apps (based on the number of customer reviews) made available by Apple Inc. through iTunes (see Table 1). Data on the use of the apps were collected through a computer program designed to “scrape” the basic information about each travel-related app (e.g., release date, version, developer, function description) and customer reviews posted for that app. By the end of July 2010, there were total 49,357 records of customer reviews under the travel category, and 37,133 records (75%) out of the total are related with the most popular one hundred travel-related apps.

A preliminary analysis of the customer reviews indicated that most reviewers commented on the specific functions and usability of the app without describing their travel experience. However, some customer reviews included discussion of the impact of the apps on their travel experience where they offered “snapshots” of how the particular app supported or enhanced their experience. Following Craig (2007), it is

argued that these reviews are “narratives” that are “the most likely medium to capture the contingencies of human experience as lived in context and over time” (p. 174). Studies in anthropology, literary theory, psychology, geography, and organizational theory have argued that the narrative is “the closest we can come to experience” (Connelly and Clandinin 1990). Recent studies have used this approach to identify the motivations of tourists in generating online word of mouth (WOM) such as self-expression, self-discovery, and solving problems (Pan, MacLaurin, and Crofts 2007; Gretzel and Yoo 2008; Bronner and de Hoog 2011) and demonstrate that online WOM (i.e., customer reviews) can provide important and useful information in describing touristic experience. Thus, it is posited that these storytelling customer reviews are extremely helpful in exploring the role of smartphone apps in mediating touristic experiences.

The screening process for identifying storytelling customer reviews began by examining all reviews (37,133 records) of the top 100 apps in terms of their content and length measured by the number of words. First, short reviews (e.g., reviews that include 100 words or fewer—a total 25,383 records) were eliminated as they focused solely on the functions of the apps and/or provided general feelings about the apps such as “love,” “amazing,” “disappointing” and provided little information about the impact of the apps on their travel experience. Second, negative reviews (with the rating 1 or 2, 10,831 records representing 29% of all the reviews) were eliminated as preliminary analysis indicated that they too focused solely on usability-related aspects of the apps. This two-step screening process resulted in 919 positive customer reviews that were considered potential storytelling customer reviews. A third step in the screening process involved two of the authors whereby they conducted a manual screening of the 919 reviews by employing the storytelling degree-of-freedom instrument (see Table 2) proposed by Woodside, Sood, and Miller (2008). Specifically, the authors read the reviews separately to look for “stories” describing the use of the apps and their role in the travel experience. It was decided that if five of the seven criteria proposed by Woodside, Sood, and Miller (2008) were met, the customer review was kept as a viable storytelling customer review. Based on this process, 202 of 919 customer reviews (22%) were classified as viable storytelling customer reviews. Table 3 summarizes the screening process used in this study. The 202 positive reviews represent approximately 0.5% of total review records (37,133) and pose a number of limitations to this study, including (1) statistical inferences cannot be drawn from the reviews and (2) the negative impacts brought by the inadequacies of the smartphones and the apps, such as weak signal and the “crash” of the apps, cannot be identified from the reviews. However, it is argued that the reviews provide sufficient insight to address the overall goal of this study, which is to understand the various ways smartphones (and their apps) mediate the touristic experience.

Table 1. The Categories of Smartphone Apps Based on the Types of Information Services

Category ID	Category Name	Definition	No. of Apps (N=100)	Percentage (%)	Examples
8	Flights information manager	Apps for searching and tracking flights	17	17	Flight Tracker; Triplt - Travel Organizer
9	Destination guides	Apps for providing various information about a particular place such as New York	15	15	Vegas Mate; NYC Way; GayCities - Your Gay City Guide
4	Online travel agency	Apps for searching and reserving transportation tickets/hotels/cars	11	11	Travelocity; Kayak flight, hotel search; Avis Reservation App
7	Facilitator	Apps for providing quick facts such as Wi-Fi spot, cheap gas stations, local times.	10	10	Cheap Gas; Wi-Fi Finder; SitOrSquat: Bathroom Finder
11	Attractions guides	Apps for providing travel tips within an attraction or a resort such as Walt Disneyland theme park	9	9	Disney World Dining; Disneyland MouseWait Social Wait Times FREE
1	Entertainment	Apps for the purpose of fun	7	7	Sodasnap Postcards; Talk Radio; Trip Journal
3	Food finder	Apps for searching restaurants	7	7	Urbanspoon; Localeats; Happy Hours
5	Language assistant	Apps for translation	7	7	Lonely Planet Mandarin Phrasebook; Translate with Voice
10	Local transportation	App for searching and reserving local transportations such as buses, subways, and taxi.	7	7	PDX Bus; Metro Paris Subway; Taxi Magic
12	Augmented reality	Apps for viewing live situations in other places through webcams	5	5	WorldView; Times Square Live; Google Earth
2	Currency converter	Apps for calculating exchange rates	3	3	ACT Currency _Universal Currency Converter
6	Tips calculator	Apps for calculating tips	2	2	Tipulator

Table 2. Storytelling Degree-of-Freedom Instrument (Adapted from Woodside, Sood, and Miller 2008, p. 116)

For each proposition below, please circle one response regarding whether the proposition matches the storytelling format:	No	Not Sure	Yes
1. Protagonist found in the customer reviews engages in actions to achieve goals?			
2. Mundane presentation of self in everyday life of a protagonist in the story			
3. Protagonist found in reviews engages in actions to achieve goals?			
4. The review presents a story that informs about conscious and/or unconscious thoughts of the protagonist and other actors?			
5. The review describes a world or personal block that does/may prevent goal attainment?			
6. The review has a beginning, middle, and end?			
7. The review presents a protagonist in the clear-cut situations?			

Content analysis was then used to analyze the selected customer reviews. Content analysis is a “research technique for the objective, systematic, and quantitative description of the manifest content of communication” (Berelson 1952, p. 18). There are two general classes of content analysis in social science: qualitative and quantitative. Qualitative content analysis refers to nonstatistical and exploratory methods

that are often an epistemology whereby “reality” is a social and cultural creation that can be accessed only by the detailed investigation (Berg 2001). Quantitative content analysis, on the other hand, refers to methods that are capable of classifying many words of text into much fewer content categories that then can be counted for their occurrences and provide statistical inferences from text populations (Weber 1990). The

Table 3. The Screening Process and Screening Criteria for the Storytelling Customer Reviews

Screening Process	Screening Criteria
All reviews	Total 37,133 customer reviews for 100 apps were “scraped” from iTunes app store.
Positive reviews	26,302 positive customer reviews with a rating of 4 or 5 were selected, as the purpose of this study was to examine how the apps can help or facilitate tourists do something.
Candidates of storytelling reviews	919 positive customer reviews were selected based on the length (≥ 100 words). Shorter positive reviews cannot reveal the touristic experience.
Storytelling reviews	202 records were identified as storytelling customer reviews based on the storytelling degree-of-freedom instrument (Woodside, Sood, and Miller 2008).

Table 4. Code Book

	Definition ^a
Information needs	
I. Functional information needs	Tourists need information to learn, increase the value of the trip, improve efficiency, reduce uncertainty, and release stress.
II. Innovation information needs	Tourists need information to be innovative, look for different things, plan new things for trips.
III. Hedonic information needs	Tourists need information to be excited, “feel” the destination, and experience the local culture and life.
IV. Aesthetic information needs	Tourists need information to imagine destinations, and form expectations
V. Social information needs	Tourists need information to give advice to others, share their experience, and be valuable for their friends.
Outcomes	Definitions (grounded from customer reviews)
a. Overall satisfaction	“a great trip,” “fantastic experience,” etc.
b. Good value	“maximize my experience,” “really add value,” etc.
c. Confidence	“no worry about anything,” “I’m the expert of . . .,” etc.
d. Easy life	“relief,” “no stress,” “easy life,” etc.
e. Life saver	“life saver,” “save my life,” “save my marriage,” etc.
f. Rich experience	“enriched my experience,” “experienced a lot,” etc.
g. Delight the trip	“delightful experience,” “delighted my trip,” etc.
h. Meet expectation	“as we planned,” “exactly as I expected,” etc.
i. Efficiency	“efficient planning,” “find my gate in a minute,” etc.
j. Show off self-esteem	“I impressed other people,” “my friends depend on my decision,” etc.
k. Inspiration for travel	“miss Disney so much,” “looking forward our next road trip,” etc.
l. Share happiness	“put miki’s smile on my mom’s face,” etc.
m. Peaceful mind	“release my nerve,” “peace in mind,” etc.
n. Visit more places	“went to many places which we don’t even know without it,” etc.

a. Adapted from Vogt and Fesenmaier (1998, p. 570).

selection of quantitative and qualitative content analysis depends on the purpose of study and styles of textual data (e.g., appropriate for summarization, similarity of different files) (Newbold, Boyd-Barrett, and Van Den Bulck 2002). It is argued that qualitative content analysis is the most appropriate approach given that the goal of this study is to explore the mediation of smartphones on the touristic experience from customer reviews (Neuendorf 2002).

Following Neuendorf (2002), the 202 qualified customer reviews were manually assessed/coded using two sets of constructs: information needs and the outcomes of the use of smartphones (see Table 4). Specifically, the development of codes assessing information needs was an iterative process whereby the initial code list was developed based on the

typology proposed by Vogt and Fesenmaier (1998) and then revised during the process of analysis. The codes related to outcomes, on the other hand, were fully grounded in the customer reviews. Importantly, the two coders worked separately for coding, and intercoder reliability was checked by calculating Krippendorff’s (1980) alpha. For the two sets of codes (information needs and outcomes), Krippendorff’s alpha is approximately 0.7 (0.77 for the information needs, and 0.69 for the outcomes), which is the threshold for intercoder reliability (Krippendorff 1980).

Finally, analysis of the co-occurrence of information needs, outcomes, and the categories of apps was examined using crosstab analysis. Then following Woodside and Chebat (2001), deconstructive analysis was used to assess

how (the nature and extent) smartphones affect the travel experience by identifying representative customer reviews from each group of information needs. Specifically, each review was deconstructed by developing an emic narrative interpretation map with the application of approach for story analysis in order to understand the protagonist's activities and emotions, and the relationship between protagonist and other key concepts within the travel experience such as other people or objects.

Results

Table 5 presents the extent to which smartphone apps address the information needs and their induced outcomes. As can be seen, all five information needs proposed by Vogt and Fesenmaier (1998) were mentioned by tourists in their reviews. As one might expect, functional information needs are most frequently described (164 cases, 81%). The second most frequently mentioned information needs relate to social information (44 cases, 22%), which is followed by hedonic information needs (33 cases, 16%), innovation information needs (18 cases, 9%), and aesthetic information needs (14 cases, 7%). Satisfaction in meeting these information needs appear to have had 14 different outcomes varying from behavior to emotional states. About one-quarter of the travelers described how the smartphone apps help them to get a "good value" out of their trips (53 cases, 26%), become more "efficient" in terms of planning and activities (40 cases, 20%), and improve their overall satisfaction (36 cases, 18%). Ten percent of the travelers indicated that the use of smartphone apps added "delight to their trips" (23 cases, 11%), helped them to "meet their expectation" (20 cases, 10%), and eliminated their stress and worries ("peaceful mind," 19 cases, 9%). Three percent to 6% of the reviews described a variety of other benefits, including providing a "rich experience" (13 cases, 6%), "inspiration for travel" (12 cases, 6%), "life saver" (11 cases, 5%), "show off self-esteem" (11 cases, 5%), "share happiness" (9 cases, 4%), "confidence" (7 cases, 3%), and "visit more places" (6 cases, 3%).

The Relationships of Smartphone Apps, Information Needs, and the Outcomes of the Use of Apps

In the following paragraphs, the relationships between apps and information needs, apps and outcomes, and information needs and outcomes are described based on the results of crosstab analysis. Table 6 presents the results of a crosstab analysis of information needs and different categories of smartphone apps. It appears that functional information needs are highly associated with the apps of "Flight Information Management" (66 cases, 32.7%) and "Attractions Guides" (36 cases, 17.8%). Innovation information needs are more likely to be associated with the apps of "Food

Table 5. Frequency of Information Needs and Outcomes

	Cases	% (N = 202)
Information needs		
I. Functional	164	81
IV. Social	44	22
III. Hedonic	33	16
II. Innovation	18	9
V. Aesthetic	14	7
Total codes frequency	273	
Outcomes		
b. Good value	53	26
i. Efficiency	40	20
a. Overall satisfaction	36	18
d. Easy life	25	12
g. Delight the trip	23	11
h. Meet expectation	20	10
m. Peaceful mind	19	9
f. Rich experience	13	6
k. Inspiration for travel	12	6
e. Life saver	11	5
j. Show off self-esteem	11	5
l. Share happiness	9	4
c. Confidence	7	3
n. Visit more places	6	3
Total codes frequency	285	

Finder" (11 cases, 4.5%) and "Destination Guides" (10 cases, 3.5%). The "Attractions Guides" apps (14 cases, 6.9%) appear to address hedonic information needs. Social information needs are more likely to be addressed by the "Flight Information Management" apps (15 cases, 7.4%) and "Attractions Guides" apps (19 cases, 9.5%). Finally, the aesthetic information needs are addressed mainly by "Attraction Guides" apps (11 cases, 5.4%).

Table 6 also presents the results of a crosstab of the outcomes induced by the satisfaction of information needs and different categories of smartphone apps. As can be seen, "Attractions Guide" apps are identified to be the most influential apps because this category of apps is highly associated with seven outcomes (highest percentage in each row) including overall satisfaction (16 cases, 7.9%), good value (20 cases, 10%), delight the trip (13 cases, 6.4%), meet expectation (9 cases, 4.5%), efficiency (7 cases, 3.5%), inspiration for travel (6 cases, 3%), and share happiness (4 cases, 2%). "Flight Information Manager" apps are more likely being able to bring easy life (19 cases, 9.5%), solutions (life saver) (5 cases, 2.5%), and peaceful mind (19 cases, 9.5%) for tourists. "Food Finder" apps can add value (good value) (8 cases, 4%) to the trips. Finally, "Destination Guides" apps are able to encourage tourists to visit more places during the trips (4 cases, 2%).

Table 6. Frequencies of Smartphone Apps, Information Needs, and the Outcomes of the Satisfaction of Information Needs

	Smartphone Apps by Categories (No. of Cases, N = 202)											
	Entertainment	Currency Converter	Food Finder	Online Travel Agency	Language Assistant	Tips Calculator	Facilitator	Flights Information Manager	Destination Guide	Local Transportation	Attractions Guide	Augmented Reality
Information needs												
I. Functional	3	1	11	6	2	1	15	66	10	12	36	1
II. Innovation			9				1		7		1	
III. Hedonic	2		6			1	1	3	4		14	2
IV. Social	3		2	1	1		1	15	1	1	19	
V. Aesthetic											11	3
Outcomes												
a. Overall satisfaction	2		5		1		2	3	5	2	16	
b. Good value	1		8	3		1	7	7	5	1	20	
c. Confidence			2				2	2		1		
d. Easy life		1	1	1			2	19			1	
e. Life saver			1					5	1	3	1	
f. Rich experience			5						3		5	
g. Delight the trip	2		3		1	1	1	1	1		13	
h. Meet expectation			1				1	7		1	9	1
i. Efficiency	1			1	1		4	22	1	3	7	
j. Show off_self-esteem			1				1	5		1	3	
k. Inspiration for travel			1						1		6	3
l. Share happiness	2		1					1	1		4	
m. Peaceful mind								19				
n. Visit more places									4		1	1

Analyses also were conducted to examine the relationship between information need and the outcome of the apps used. Figure 2 displays the association between information needs and the outcomes where the co-occurrences are based on Jaccard (Cheetham and Hazel 1969). Specifically, Jaccard's coefficient is computed from a fourfold table as $a/(a+b+c)$, where a represents cases where both items occur, and b and c represent cases where one item is found but not the other. The results of this analysis indicates that the strongest association is between "Aesthetic information needs" and "Inspiration for travel," followed by "Innovation information needs" and "Rich experience"; "Functional information needs" and "Good value"; "Hedonic information needs" and "Delight the trip"; and "Social information needs" and "Peaceful mind." In addition, satisfaction related to social information needs is associated with efficiency in the trip; satisfaction related to functional information needs is associated with improving overall satisfaction, and satisfaction related to hedonic information needs may induce shared happiness. Last, satisfaction related to innovation information needs encourage tourists to visit more places.

Narrative Interpretation of the Touristic Experience Influenced by Smartphones

The stories provided by tourists can provide for a vivid understanding on the mediation mechanisms of smartphones in the touristic experience. This section presents a series of

travelers' reviews that reflect different ways (scenarios) travelers have used the respective applications and their impact on their travel experiences. The scenarios are organized based on the information needs typology established in the previous section. In particular, tourists' stories in each scenario were selected considering the scenario's representativeness of other stories that reveal the relationships between the apps, information needs in that how they are addressed, and the outcomes of the use. As a result, two or three representative customer reviews are deconstructed for each scenario into emic narrative interpretation maps to display the touristic experience following Woodside and Chebat (2001) and Woodside, Cruickshank, and Dehuang (2007).

Scenario I: Good Value and Efficiency

It is usually difficult for tourists to predict all of the situations that may happen in the travel. Thus, tourists may not be able to make plans to cope with unexpected situations and further to maximize the benefits of their travel. For example, accidents during a trip can easily ruin the whole vacation; also, the unpredictable situations may limit tourists in following the original plan. Therefore, during the trip tourists may need extra information in order to cope with the unpredictable situations. Thus, instant access to information support may be important in addressing emergent information needs. Importantly, meeting with these information needs may improve the efficiency in the travel process and maximize

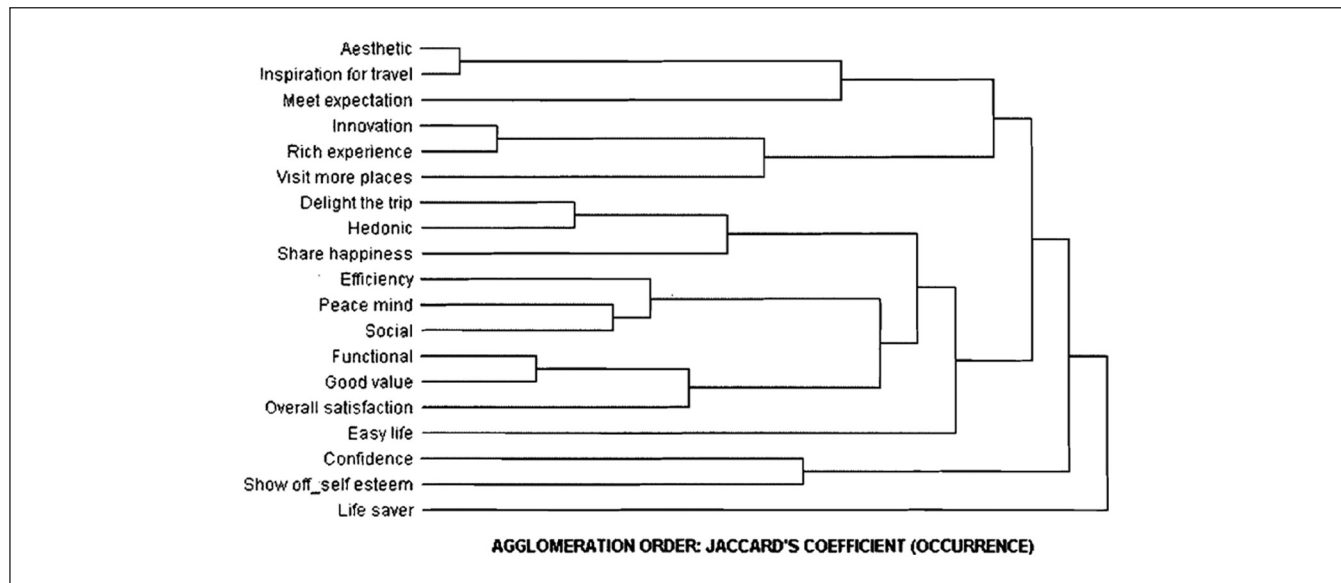


Figure 2. Clustering of information needs and outcomes

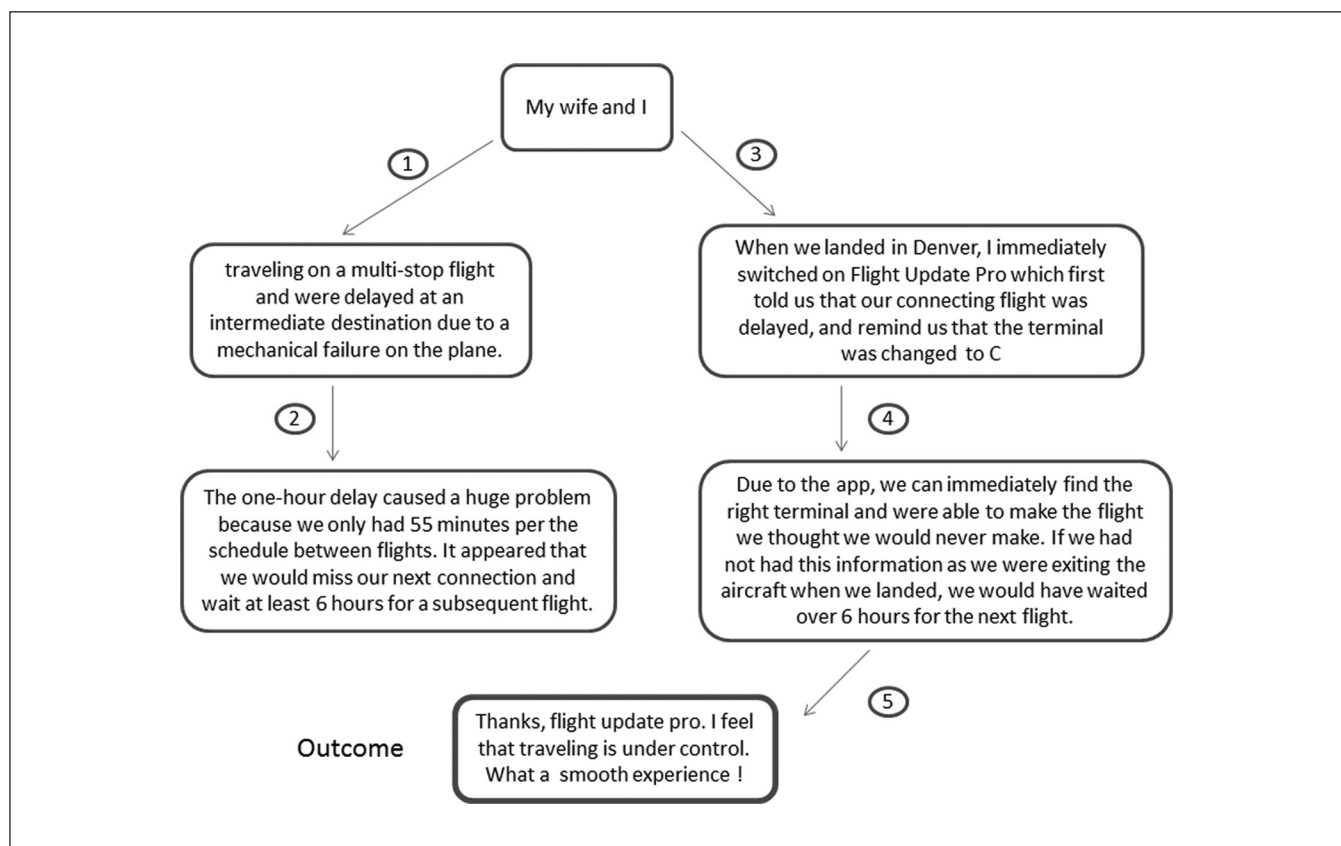


Figure 3. Tourist's story about the smartphone app "Flight Update Pro" and "Efficiency"

the value of the trip. In the following two scenarios, smartphones mediated touristic experience by awarding tourists "a smooth travel experience" and "maximizing the travel experience."

The scenario represents about 44% (90 cases) of the storytelling customer reviews ($N = 202$) which have different contexts but reflect the nature of using the apps to add value. Figure 3, for example, presents a story about how the app

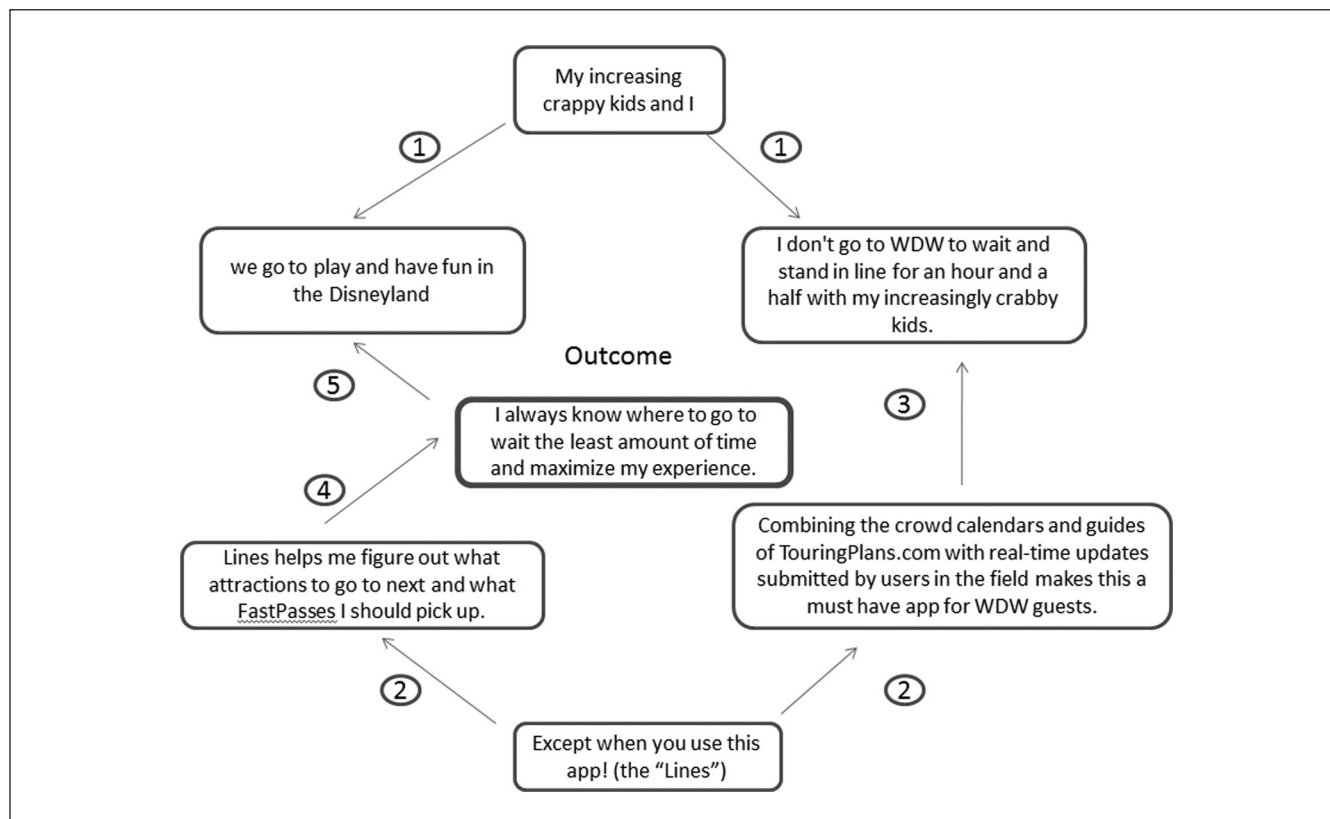


Figure 4. Tourist's story about the smartphone app "Lines" and "Good Value"

"Flight update pro" facilitated the protagonists (Case 226) in the multiflight transition process. The protagonist was traveling with his wife on a multistop flight and they were delayed at an intermediate destination, which may have led to missing a subsequent flight if they could not find the connecting gate in the shortest time. With the app "Flight update pro," the protagonist was able to "immediately find the right terminal," avoid "waiting over 6 hours for the next flight." As a result, the protagonist felt "traveling is under control" and had "a smooth experience." Figure 4 presents a second story describing how the app "Lines" helped the protagonist (Case 69) to do instant on-site planning while visiting the Walt Disney World (WDW). As can be seen, the app "Lines" helped the traveler to get updated information about the waiting time for each ride in that he or she could figure out the most efficient plan in the park. The result is that the protagonists felt that they had a "valued" trip.

Scenario II: Visiting More Places and Having Rich Experiences

Tourists like "surprises" in the travel process because unexpected activities may bring extra happiness to tourists; therefore, tourists keep looking for new and adventurous experiences (Snepenger 1987). To reduce the risk embedded in adventurous experiences, tourists need information to

support decision making; as illustrated in the following scenario, smartphones were used to change activities and have adventures by providing instant information support. These storytelling customer reviews included 11 cases (5% of total 202 cases) whereby tourists describe their experience in which the smartphone apps helped them to learn, replan, and visit new places and/or attractions. And as illustrated in Figure 5, the traveler (Case 577) extended his or her road trip because the "Roadside America" app lead to "places I might never have experienced or heard." As a result, the protagonist felt that "the app enhanced my trip a thousand fold from what it would have been like with only maps and travel guides."

The touristic experience also may be enriched by the availability of a variety of options. Because of the intangibility of travel products, tourists largely experience the destination at the time of consumption and may change their plan if it does not match what is expected. Smartphones help by providing suggestions based on tourists' preferences and assisting tourists to personalize the search results. For example, Figure 6 is a story about a frequent traveler. The protagonist (Case 534) was tired of the local food in the destination he/she visited several times (within a month), writing "I can only handle so much 'Paula Dean'-type choices" and wanted to have a different dining experience. The "Open table" app helped the traveler find "the places the

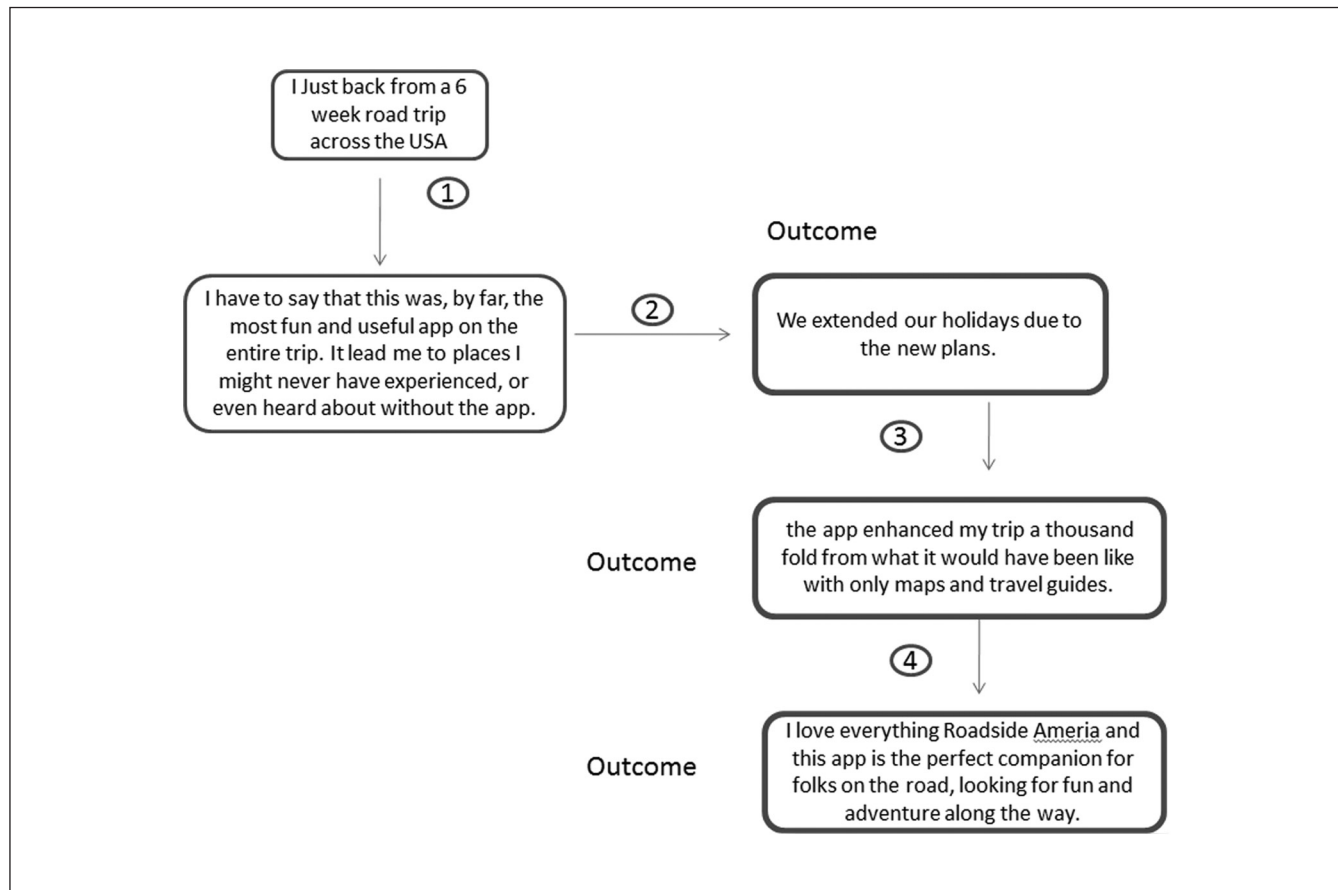


Figure 5. Tourist's story about the smartphone app "Roadside America" and "Visit More Places"

locals did not know existed," which resulted in a feeling that the app "delights the boring trips."

Scenario III: A Delightful Trip and Higher Satisfaction

Tourists are hedonic pleasure seekers for fun, enjoyment, amusement, and sensory stimulation. The information services provided by smartphones also appear to elicit tourists' hedonic pleasures by providing novel information. "Delightful trip" and "Higher satisfaction" are two outcomes of using the apps (10%, 21 cases of total 202 cases). For example, the "Air Traffic Control" app helps tourists to listen to the conversation among crews and aviation control staff and eases the boredom of tourists in the waiting time in the airport. As illustrated in Figure 7, the protagonist (Case 476) indicated that the "Air Traffic Control" app can make the long and weary waiting time at the airport fun and enjoyable. Also, the protagonist perceived that the app is "both entertainment and comfort" for a nervous flyer because he/she can know that "the plane and the ground are working together." In addition, the storytelling customer reviews show that mobile apps can help tourists recognize the hedonic aspects of

experience, which leads to higher overall satisfaction. Figure 8 presents the story of the app that can award tourists more fun when they were able to make postcards and share with friends right after taking photographs on-the-spot. The protagonist (Case 568) indicated that she would like to share her travel experience with family and friends in the form of postcards, but she hated to shop for local postcards and send via the post office. The "Postman" app enabled her to share the "fabulous sunset" with her husband in the format of a self-made postcard, and as a result, the protagonist really enjoyed the trip (particularly the sunset) because of the unique experience cocreated by the app.

Scenario IV: Sharing Happiness, Showing Off, and a Peaceful Mind

Many of the storytelling customer reviews reveal that smartphones mediate the travel experience by enabling tourists to connect with not only families but also new friends during the trip. The following three scenarios illustrate the outcomes of smartphones using social media apps and represent about 13% of the total storytelling customer reviews (26 cases). By providing a platform for an online community, smartphones

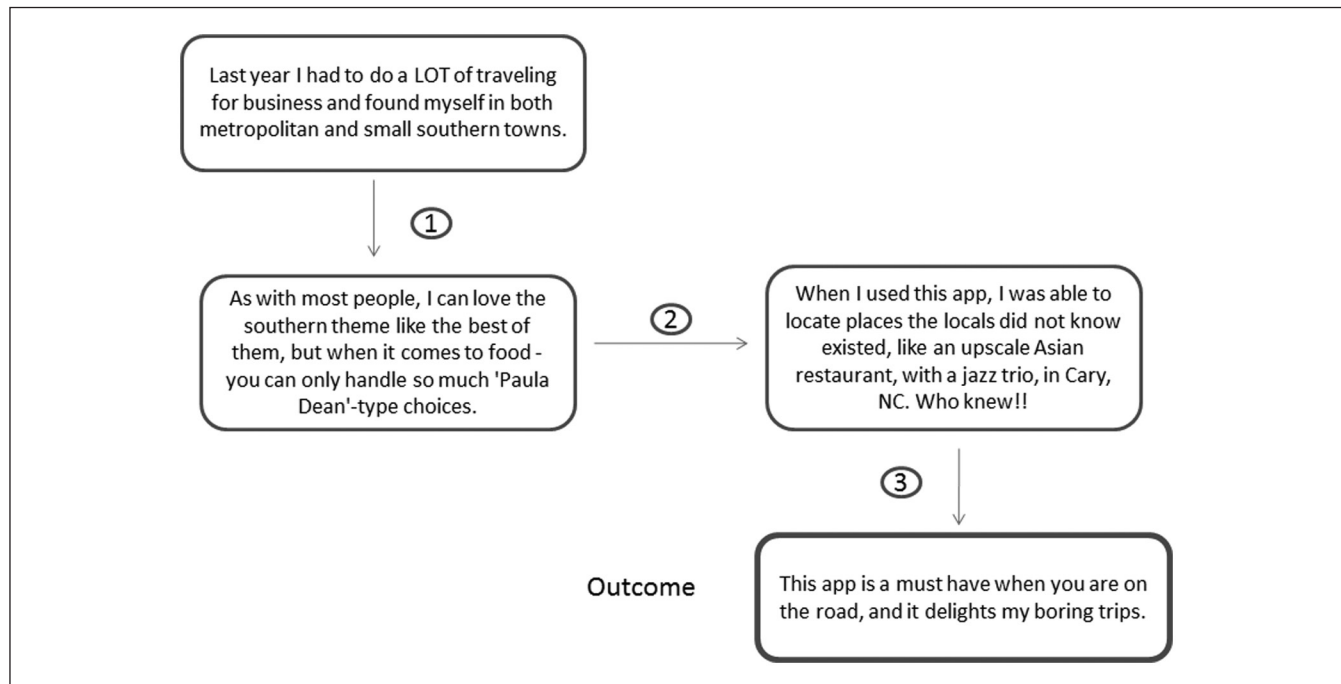


Figure 6. Tourist's story about the smartphone app "Open Table" and "Rich Experience"

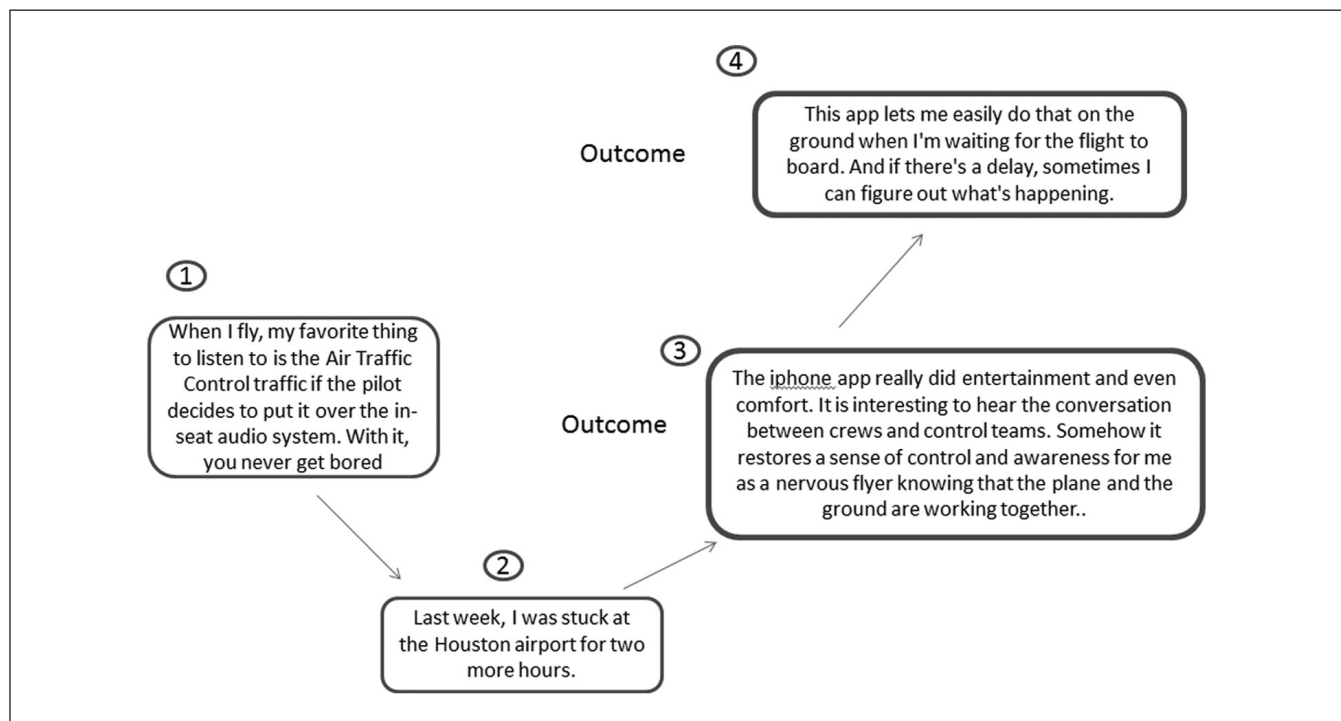


Figure 7. Tourist's story about the smartphone app "Air Traffic Control" and "Delightful Trip"

can connect tourists who are visiting the same destination/attraction and looking for similar information. Such "portable" online communities enrich the travel experience in many ways. Figure 9 illustrates the influence of mobile online

community where the protagonist (Case 146) visited the Disneyland with his or her friends and they wanted more people to join their group for fun. With the "MouseWait" app, they got connected with other tourists in the park at that

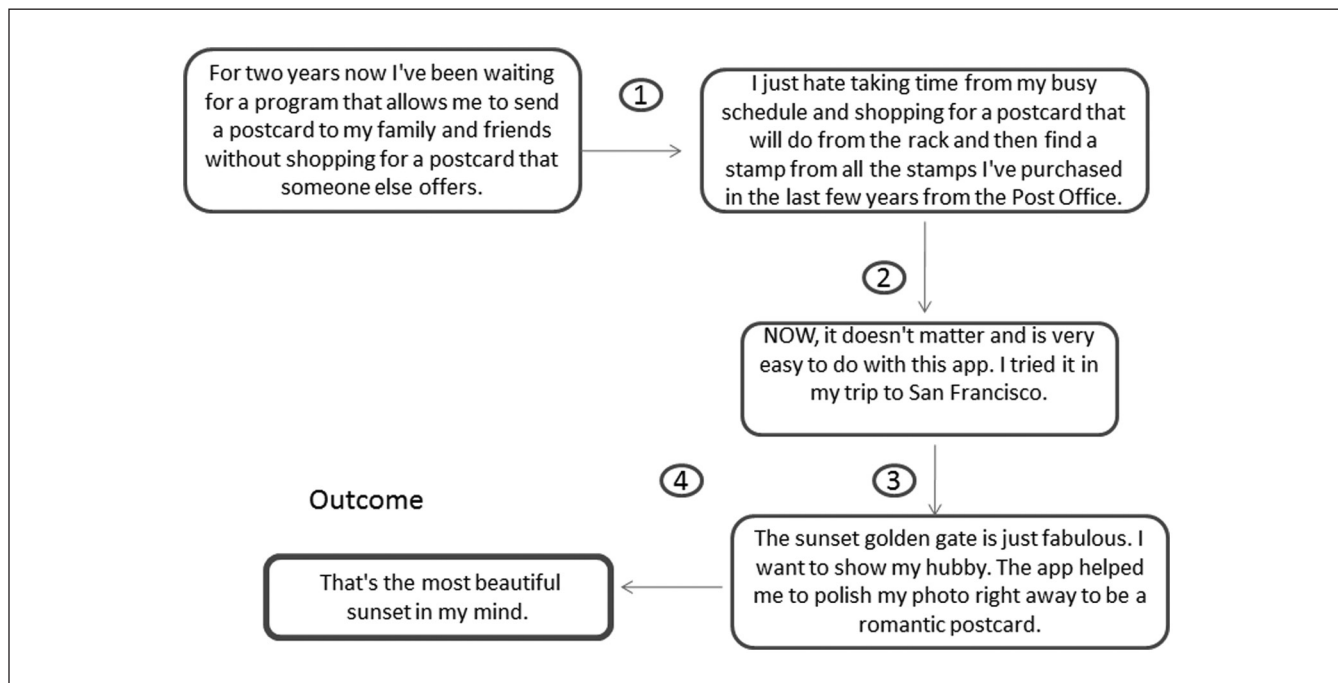


Figure 8. Tourist's story about the smartphone app "Postman" and "Higher Satisfaction"

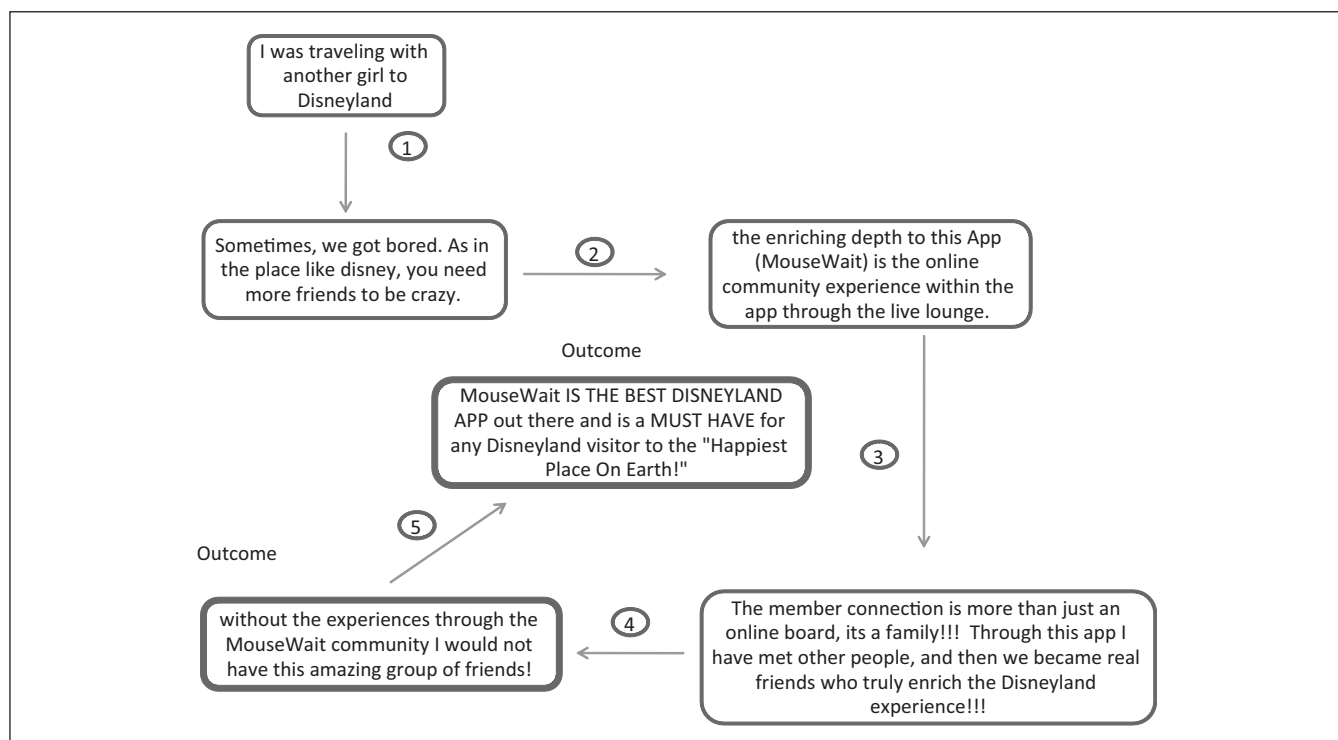


Figure 9. Tourist's story about the smartphone app "MouseWait" and "Share Happiness"

time and they finally met in the park to have fun together. The protagonist had a better experience in Disneyland than he or she expected as he or she felt that the helpful app made Disneyland "the happiest place on earth."

Also, the advantages brought by smartphones in instant information accessing can make tourists feel special in a group. For example, the protagonist (Case 330) had a delayed flight when he or she was in a trip from Portland coming

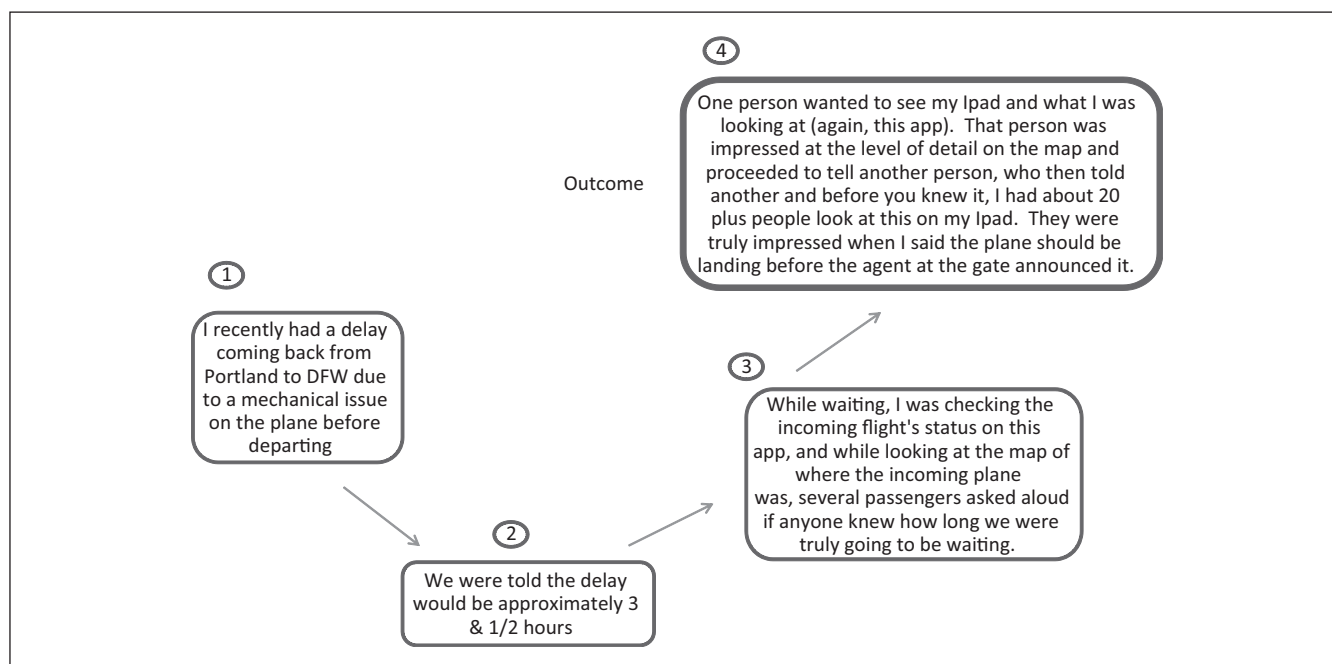


Figure 10. Tourist's story about the smartphone app "Flight Status" and "Show Off"

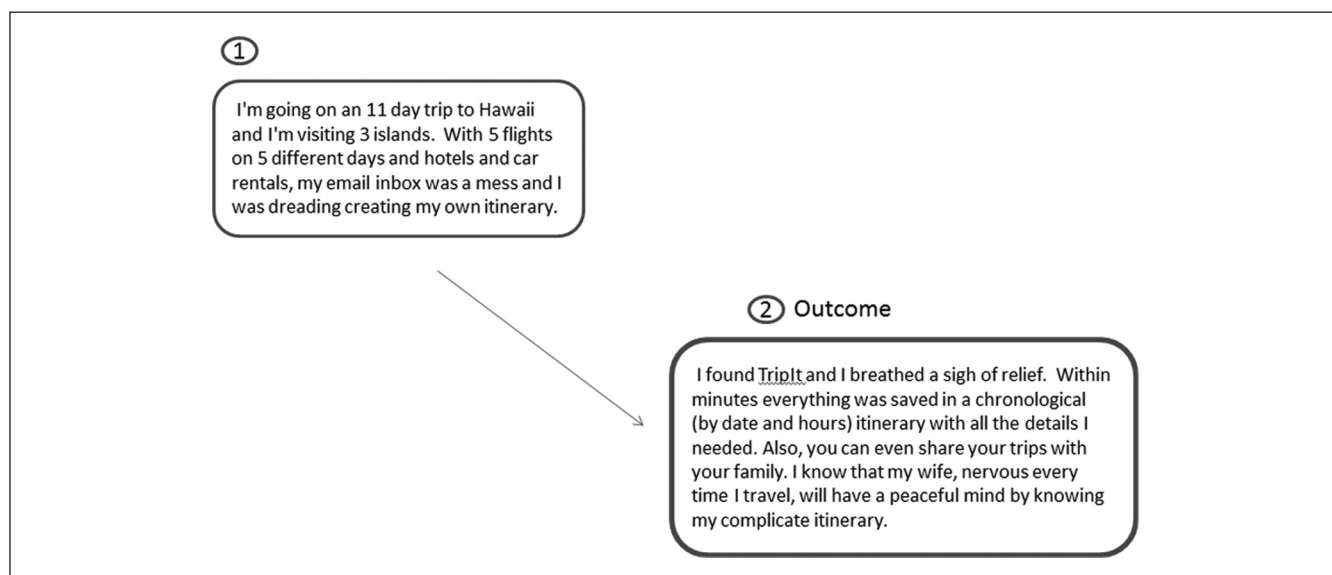


Figure 11. Tourist's story about the smartphone app "Triplt" and "Peaceful Mind"

back to his or her home (Figure 10). When he or she was using the app "Flight Status" to check the incoming plane on the map, other passengers waiting for the same flight were "impressed at the level of detail on the map." It turned out that "about 20 plus people look at this (the app) on my iPad." The protagonist was very proud that he or she could say that "the plane should be landing before the agent at the gate [was] announcing it" and that the protagonist's trip was "improved" by the use of smartphone because the appreciation from others made him or her feel "really good."

In addition, the ease of information sharing enabled by the smartphone apps helps tourists to keep their families, friends, and even themselves informed. As a result, the updated information eliminates the worries and concerns caused by the uncertainty of travel. Figure 11 presents a story (Case 723) of a traveler who used the app "Triplt" to manage his complicated travel schedules and share the information with his wife, which resulted in "a sigh of relief" for this tourist because he wouldn't be confused by the itineraries and it led to "a peaceful mind" for his wife because she could know her husband's schedules.

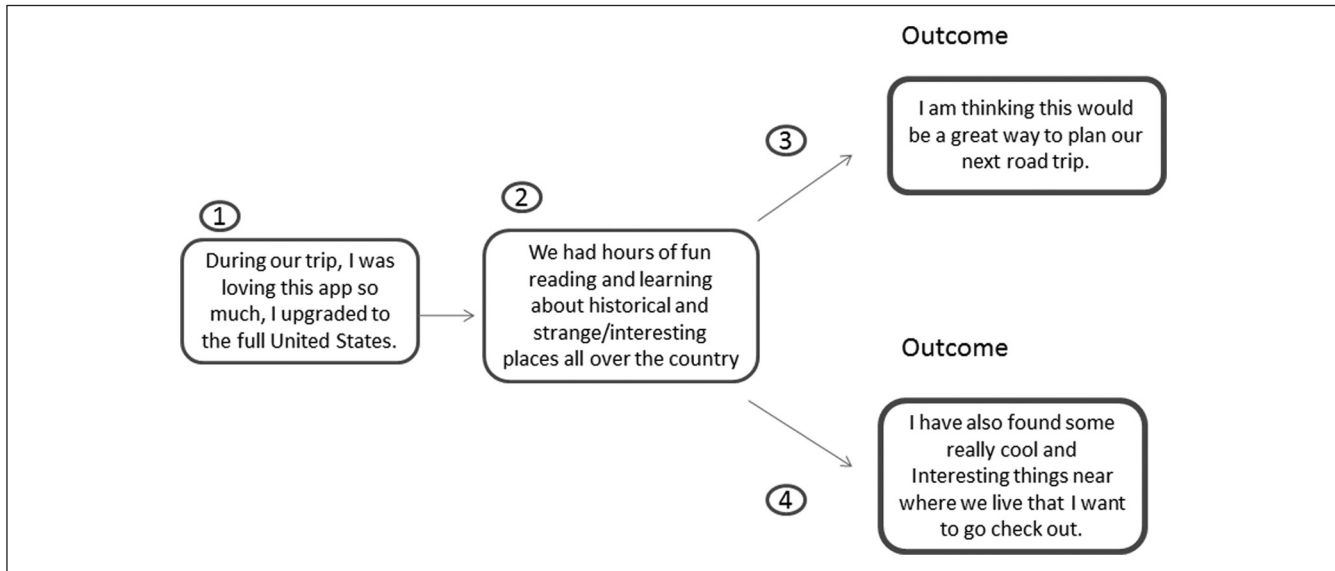


Figure 12. [CE: Text citation for Figure 12 is not provided. Please check.] Tourist's story about the smartphone app "American Treasures" and "Inspiration for Travel"

Scenario V: Inspiration for Travel

Aesthetic consumption is an interesting part of the touristic experience (Vogt and Fesenmaier 1998). Unlike functional needs aesthetic needs are self-evoked (Hirschman 1983). Smartphone apps can provide information to help the construction of the imagery of destinations and attractions, and in turn, inspire people to travel more. In this study, 5% of the total storytelling customer reviews (9 cases) somehow reflect this benefit of smartphone use. For example, Figure 11 presents a story about a person using the "American Treasures" app and the inspiration it provided for the protagonist to travel. Specifically, the protagonist used the app to get to learn about destinations near the places visited which "inspired" new plans to visit the area.

Discussion

Customer reviews of smartphone apps provide "snapshots" of the touristic experience, and analysis of these reviews reveals how their use mediates the touristic experience by changing behavior and emotional states. In particular, the smartphone apps considered in this study enable tourists to cope with unexpected situations ("life saver," "easy life," "peaceful mind") and to complete travel activities more efficiently and effectively ("good value," "efficiency," "meet expectations"). For example, the stories show how Disneyland apps indicate that travelers rely heavily on the respective apps to coordinate and interpret their activities (i.e., experiences); also, some of the smartphone apps helped create surprises ("delight the trip," "visit more places"), provided excitement ("rich experience"), encouraged imagination ("inspiration for travel"), and even gave a traveler

a sense of achievement ("confidence," "show off self-esteem"). These findings demonstrate that smartphones enable travelers to go beyond the "hermeneutic circle" to become more creative and spontaneous, which in turn leads to greater satisfaction. Finally, the results of the study indicate that the "mediated gaze" as described by Lagerkvist (2008, p. 351) may be intensified and extended when travelers communicate with families and friends and other people.

These results clearly demonstrate that smartphones are an extremely powerful channel to communicate with existing and potential visitors. It was only 10 years ago when destination marketing organizations were at the cusp of the Internet revolution (Gretzel, Yuan, and Fesenmaier 2010), and now it appears that they face a series of new challenges because the increasing use of smartphones will become an even more dominant force shaping visitor behavior. As such, it is posited that destination marketing organizations as facilitators of local tourism businesses should now take leadership by integrating marketing programs across channels, and adopting new business models that exploit the strengths of the mobile environment. Indeed, it appears that destination marketing organizations should "reinvent" themselves whereby they deemphasize traditional (including online marketing activities) to focus on mobile communications. As shown in this study, the ubiquity and strong computational capability of smartphones facilitates and encourages travelers to use location-based apps to identify/choose restaurants, attractions, and shops spontaneously rather than plan in advance or rely on more traditional information sources such as concierges or even friends and family. As such, new marketing strategies should be developed whereby destination marketing organizations (and travel-related businesses) embed promotions into the search results of the apps providing such

services and/or adopt coupon marketing programs offered through online firms such as Groupon.com. In addition, the finding that smartphones are seen as a portable platform for social communities that encourages/facilitates interaction among tourists demonstrates that such instant interaction can have a profound effect on touristic experience; destination marketing organizations (and other tourism businesses), therefore, need to support forums that enable visitors to communicate with each other and members of the network by matching the form and texture of the respective forum (Kozinets et al. 2010).

Importantly, Turkle (2011), among others (Baym 2010; Gretzel 2010; Yoo 2010), suggests that the increasing penetration of mobile devices in people's daily life will have a profound influence on how people think and behave. Following from this work and the findings of this study, it is posited that these changes in daily life will lead to significant changes in travel behavior as the unique features of the travel context (i.e., out of home, search for new experiences, and unpredictability) may encourage travelers to rely even more on mobile devices. The implications of these changes are significant and suggest a number of important streams for future research. First, research should focus on the influence of mobile devices on travel planning. Because of the ubiquity of mobile devices and the incredible depth and breadth of innovation of mobile apps, it is argued that travelers may change information channels by substituting other digital devices (i.e., video recorder, mp3 players, etc.), and as a result, travelers' decision-making schema may change dramatically. For example, it is expected that the choice of hotels, restaurants, and attractions will be influenced extensively by use of mobile apps. The app "Urbanspoon," for example, may substantially change tourists' choice of restaurant because of its innovative "shake" function, which encourages surprise; similarly, the music station "Pandora Radio" uses a sophisticated recommendation system to encourage listeners to discover new music. A second area of research focuses on the interaction of travelers' use of information services available on mobile devices, and the development of information services based on the argument that travelers and smartphone information services are shaping each other's landscape with the increasing penetration of smartphone information services in people's lives (Pan et al. 2010). As such, studies should examine the extent to which the smartphone/apps become an extension of oneself in supporting ongoing search and communication within the "community." The third area of research focuses on the interaction among tourists, the destinations they visit, and the people that travelers share experience with while en route. Following Kozinets et al. (2010), it is argued that it is essential that we understand how mobile social networks encourage instant sharing of travelers' stories, and how the instant feedback supported within these networks may lead to new activities in the destination as well as the "reinterpretation" of the touristic experience. It is clear that we are only beginning to

understand how mobile technology shapes tourism, both from traveler and industry perspectives; it is hoped that this article provides a foundation for appreciating the impact that the mobile Internet is having on travelers and the travel industry.

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